

Everything but Programming; Investigating Academics' Perceptions of Embedded Soft Skills in Computer Science Undergraduate Education

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Student employability is a key goal of a computer science undergraduate education. A soft skills gap has previously been reported between employer requirements and the skills graduates offer, suggesting that educators are inadequately preparing students for their future careers. It is important to identify the links between educators and the materials they claim to teach as it offers insight into how non-technical aspects of software engineering are promoted. We report on two studies where we first explore the staff perceptions of embedded soft skills in five computer science undergraduate courses, before identifying soft skill presence in curricula across eight UK universities. A multi-site interview with educators identified core skills of *critical thinking*, *communication*, and *teamwork* being included in curricula for student employability. Staff believe students experience a temporal delay between being introduced to skills and *actually* valuing them. In the second study, we mined publicly-available course and module information, and then analysed non-technical skill references. Soft skills are commonly found in proximity to other soft skills, suggesting they are taught or assessed together. *Software engineering* was seen to be closely linked to *teamwork* and *communication*, emphasising it is taught as a social enterprise. Taking these two studies together, educators show a close alignment to curricula, and the skills valued by higher education institutions reflect the skills valued in software engineering industries, suggesting the skill gap is the result of student misconceptions.

CCS Concepts: • **Security and privacy** → **Social aspects of security and privacy**; • **Human-centered computing** → *Empirical studies in HCI*;

Additional Key Words and Phrases: soft skills, graduate employability, student misconceptions, curriculum analysis

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1 INTRODUCTION

Graduate employability is a key outcome of an undergraduate education. Recent Computer Science (CS) graduates face intense competition when seeking employment, and in 2023, CS undergraduate applicants in the UK increased by nearly 10% [7], suggesting an ever-growing body of prospective employees vying for the same job positions. Consequently, students want their education to provide them with the key skills required to stand out from other job applicants. Students are reliant on staff to teach them the necessary skills [30], but student misconceptions can restrict their ability to fully engage with the teachings, leading to incorrect value being assigned to technical and non-technical skills. Following graduation, students are then perceived to not possess the soft skill competencies required by employers, resulting in a soft skills gap between education and employment [4]. From both the student and employer perspective, it is the educator who is responsible for developing these soft skills, and so an exploration of their perceptions is beneficial to further illuminate the gap, and surface important issues around soft skill development.

Soft skills are the domain-agnostic, non-technical skills that determine the success of technical skill application [34], and they are emphasised as being important to the health of software engineering [12]. Despite this, students' expectations do not typically accommodate soft skills, which leads them to undervalue educator efforts to nurture these skills. This results in the soft skills gap between employer expectations and what graduates offer [36]. The situation raises important questions that deserve answers. How do staff design and implement educational materials to reduce this soft skills gap? How do staff handle the tensions between what students want on, and what they need (as determined by prospective employers)? In this empirical project, we report on two complementary studies, where we evidence how staff address and resolve this tension, as well as extract soft skills from curricula material to understand how skills are embedded across the educational pathway.

Employers, students, and educators comprise a stakeholder triad, each group invested in the soft skill development process but with different motivations. Employers are motivated to recruit graduates who offer the necessary skills, because they directly experience the consequences of employees not possessing the relevant skills [44]. Students are motivated to gain the necessary experience and expertise required to secure employment [54], and educators are motivated to teach the necessary soft skills because graduate employability is an important metric, and not least, because they want to ensure future generations of software engineers possess the skills required to make an impact. Out of the three stakeholder groups, only the student population is expected to actively change and develop expertise in the soft skills being taught. Employers expect this change to occur during education [47] because educational environments are suitably-placed to allow students to test and develop their own ideas and understandings of soft skills. Despite the different motivations, they are insufficient to close the skill gap. While students are those who actively need to develop these skills, educators are directly responsible for cultivating the necessary environment for soft skill development, and it is this that we are interested in: the educator perspective.

Educators are not unsupported when it comes to deciding which soft skills to include in their curriculum. The SE2014 (Software Engineering 2014 [6]) provides a framework for CS curricula design that includes key soft skills to be taught. SE2014 is built upon the Software Engineering Body of Knowledge (SWEBOK [8]), which represents the skills needed for software engineering. The fact that SE214 is derived from SWEBOK suggests that a strong alignment between what employers want and what educators teach, but how staff interpret the guidelines and then implement then is not necessarily homogeneous, and it is this that we seek to understand.

Recognising that the closing of the soft skills gap is not a unilaterally-sided task, educators must work to present the necessary skills, but employers need to recognise the tension between student wants and student needs. To this end, we need to evidence the importance of soft skills, recognise the need for learning and training opportunities, and understand how to direct the triad of stakeholders towards a common position. Identifying the embedded skills is essential as it reflects the human-intensive nature of software engineering [12]. Our research was motivated to address the soft skills gap, by exploring the educator stakeholder perspective. We identified three preregistered research questions to guide the study:

1. How do academic staff understand how soft skills are taught in their core modules?
2. What importance do staff ascribe to certain soft skills, in comparison to other skills?
3. How do important soft skills align with the information available in online course and module information?

We answer these questions in this two-study project. The first study reports on a multi-site interview study regarding educators’ perceptions on soft skills within their module and CS course. The second study deploys a content analysis to extract references to soft skills from publicly-available curriculum content. Taken together, the findings offer a unique perspective into how soft skills are embedded across CS teaching, and how students are perceived to receive these skills.

2 RELATED WORK

The employer-student-educator triad helps to contextualise the issues around the topic of soft skills in CS education. In this section, I address relevant literature on the perspectives held by prospective employers, students, and academic staff.

2.1 Future Employer Perceptions

Employers typically prioritise soft skills [50], in part because technical skills are perceived as easier to teach to new employees [40]. By reducing the on-boarding time for new employees, they can meaningfully contribute to the company quicker, which benefits the company as it frees up resources otherwise required for training. Despite this, graduates are reportedly approaching companies without the required soft skills making them a less attractive hire, and particular deficiencies are seen for soft skills such as communication and group working [31].

Employers explicitly request soft skills in job adverts [22], emphasising that these are expected from prospective employees. Employers typically display a homogeneous view on the relevant soft skills, just with small differences in the order of importance [4]. Two primary soft skills are teamwork and communication [2, 26, 35, 42, 45]. Collaborative skills are also seen to be tied into contemporary methods of working, such

as Agile approaches [49]. Problem solving is also reported as being essential for various roles [2, 21, 41]. These skills are repeatedly identified as being essential for software engineering.

Employers are seen to blame educators for the soft skills gap, suggesting that they are failing to make students aware of the importance of soft skills [36], and others emphasise the need for educators to ensure curricula include adequate soft skills [47]. This occurs despite the SE2014 being in place which suggests that the support for curriculum design is derived from software engineering practitioners. Despite this, Akdur [4] reported that graduates are perceived not to possess the required skills when entering employment.

2.2 Student and Graduate Perceptions

To maximise impact and effectiveness, students must buy-in to the relevance of soft skill development, as they are the only stakeholders who are required to develop these skills. Broadly speaking, students are seen to hold misconceptions about soft skills, and typically prefer to prioritise technical knowledge acquisition over soft skill development [33], which is particularly evident for students with technical career aspirations [32]. This misalignment of priority skills occurs despite evidence for collaborative skills improving software quality in student work [39].

Students also conflate enjoyment with value during tasks [37] and so if students have a negative experience with an activity or project that they associate with soft skills, they will attach a lower value to these soft skills, in favour of tasks and activities that they see as more enjoyable. This may feed into their conception that technical skills are more important [33], for example programming tasks can be typically autonomous alleviating the need for collaboration with others, and it can be satisfying to solve problems through code. Students are seen to value distinctly different skills than those reported as desirable by employers. Students focus more on networking and other skills that they perceive as key to securing a job [51], rather than developing the skills asked for by employers.

Gathering data from graduates can be beneficial as they represent a population of ex-students who have experienced the educational process. CS graduates recognise the value of soft skills in their careers, emphasising the value of social and cognitive skills [11, 56]. Valued skills are not all developed during education, with group-working skills reported as being developed in employment [34], highlighting that a CS education may not be as effective as intended. Graduates have expressed concerns over their preparedness for collaborating in employment [16], indicating they do not perceive educators to have successfully carried out their role. From a contrasting perspective, when asked about the development of professional skills (which includes critical thinking, ethics, and group working), graduates report that they possess skills needed for employment [15].

2.3 The Educator's Role

Students trust educators to teach them the skills needed for employment [30], and subsequently educators need to ensure that their courses provided opportunities for students to develop soft skills, ideally over an extended period to allow for a gradual development process [28]. Amidst suggestions from industry that educators need to be teaching these skills [48], it benefits everyone to understand what soft skills are being taught and how they are embedded within curricula.

Examinations of curricula and modules can expose the soft skills that are prioritised within computer science and software engineering, with a systematic review identifying self-reflection, conflict resolution,

communication, and teamwork as the most taught skills [27], which were reflected in a further examination of soft skill-specific modules, identifying teamwork, ethics, communication, and presentation skills [25]. Others identified ethics and analytical thinking styles as being valuable within curricula [18].

Educator perceptions influence student perceptions, and students converge towards staff over time, resulting in increased soft skill valuation [24], with late-stage students demonstrating closer alignment to their educators than early-stage students do [10]. In general, those who are further through their education align more closely with the views of educators [38]. These findings highlight that students do gradually develop a better representation of the role of soft skills, but it is still limited, and total agreement is not achieved.

3 STUDY 1

Study 1 is a multi-site interview study with 12 educators across five institutions. This study offers insight into staff perceptions of embedded soft skills within core undergraduate modules.

3.1 Methodology

The study was preregistered on the Open Science Framework, lodging details on the study design, data collection strategies, and analytic plans. The preregistration can be accessed at https://osf.io/9yt35/?view_only=d8a5ff2663714a11a2ea250a59bfef81. Data, analysis scripts, and supplementary materials are found at https://osf.io/25ecz/?view_only=917e6de233484d59abaac243647c89a2.

3.1.1 Participants. Interviews were conducted with 12 staff members from five universities in the UK N8 research group (Durham, Lancaster, Leeds, Liverpool, Manchester, Newcastle, Sheffield, and York). This university group was chosen as they are all research-intensive institutions in the North of England, highlighting a shared commitment to research excellence that feeds into teaching. Educators teaching core modules for the Computer Science (UCAS code G400) undergraduate course were invited to participate via email. Participants from five institutions responded. Four from institution A (11 contacted), two from B (13 contacted), two from C (11 contacted), three from D (17 contacted), and one from E (12 contacted). See Table 1 for further details. For anonymity, participants were assigned random identifiers denoting institution using letters A through E, and a number (e.g., A1, B3) to distinguish individuals. No interviewees were personally known to the interviewer.

The invite explained the research and purpose of the contact. Those who responded were sent further information and a link where they could schedule an interview. Three educators expressed interest but did not respond further. Four declined, stating their module did not include soft skills. All participants provided informed consent. The Faculty Ethics Committee approved the research.

3.1.2 Methodology. A semi-structured interview was deployed, and the interview guide is in Appendix A. Beforehand, the interviewer made notes regarding participants' modules, department information, and any information offered during email exchanges. Each interview started with the interviewer describing the project, allowing participants to ask questions, and confirming they consented to the interview being recorded for transcription.

The first question, "tell me about the undergraduate module(s) that you teach", was designed to engage participants in a narrative mindset and provided information to direct the interview further. Questions were

Table 1. Relevant participant demographics, including their position within the department and the delivery year of core modules.

Interviewee	Position	Module(s) Year
A1	Lecturer	1
A2	Lecturer	1
A3	Reader	2
A4	Professor	1
B1	Senior Lecturer	2
B2	Lecturer	2
C1	Senior Lecturer	1
C2	Senior Lecturer	1, 3
D1	Lecturer	1
D2	Senior Teaching Fellow	1
D3	Teaching Fellow	1
E1	Senior University Teacher	1

based on the interview guide and interviewer notes. The first author conducted all interviews via Microsoft Teams, which provided automatic transcriptions and were kept to one hour. Transcriptions were validated against the original recordings and were anonymised. Transcriptions were sent to interviewees for content approval and consent to transcripts being shared under open science practices. Approval was sought to ensure staff were happy with the anonymisation carried out. Minor changes were made following feedback, which only improved interviewee anonymisation.

3.1.3 Analysis. Data were subject to a thematic analysis following the five stages: code familiarisation, code generation, theme generation, theme review, and theme naming [9]. The first author familiarised themselves via the transcription validation process and multiple readings. Initial notes were made during readings, and preliminary codes were developed. This process was iterated until no new quotes were added and no additional codes were generated. This process ended when further reading generated no substantial changes. Codes were then reviewed, which included merging or splitting codes where necessary. Themes were then formed by grouping the codes. Themes were reviewed through authors’ discussion and assessed for independence, coherency, and distinctiveness. Finally, themes were given relevant names. Transcripts were coded in Taguette, an open-source qualitative analysis tool [46]. The codebook and the tagged dataset are available in the OSF repository.

3.2 Results

Two key themes that contribute to answering the research questions were identified. The first theme of *Valued Soft Skills* relates to soft skills identified as components of teaching materials. The second theme of *Student Perceptions* relates to how educators understand the student experience. The quotes have been edited for clarity and coherence, but the original intent is maintained. The use of quotes is selective rather than exhaustive and represents staff perceptions.

3.2.1 Valued Soft Skills. The soft skills mentioned can be split into three skillsets: Cognitive, Social, and Professional. Cognitive skills relate to internal mental processes for reasoning and applying technical

knowledge. Social skills are those involved in interpersonal communication and are required to facilitate successful interactions with others. Professional skills are required for high performance and functioning in a work environment, in the context of this research, they are distinct from social and cognitive skills.

Cognitive. Educators emphasised the cognitive development opportunities that students are exposed to throughout the course. Many of these opportunities include tasks involving programming and other related activities, such as code comprehension or code writing. One exception to this general rule was educators deploying cognitive puzzles, such as sudoku, to teach logic-based reasoning. Removing the coding element ensures students develop the cognitive foundations needed for programming tasks. These puzzle-solving sessions preceded programming tasks and offered students opportunities to deploy the logical processes and programmatic approach required to program successfully.

“The aim over a series of seven or eight [sessions] is to teach all the skills of programming without ever writing a line of code. That’s our overall aim of these workshops, and they’re all generally built around logic puzzles.” – A2

Other opportunities for students to develop mental models needed for more complex tasks involved using low-level languages such as Assembly, which was used for its machine-readable syntax; by exposing students to programming languages with minimal abstraction, they can develop their internal representation of how programming works. Almost opposite to using non-programming tasks to teach logic, low-level languages can be used to teach about how programming works without requiring knowledge of complex functions. Doing so gives students the mental models to use when working with high-level languages that abstracts many low-level language behaviours.

“They’re looking at code that they’ll never use again, but it’s giving them the mental foundation so that when they’re writing in a higher level language, they can refer back” – C2

Students want to be taught fashionable languages, but educators stress the need to teach languages that offer pedagogical advantages (such as Assembly) because they help students develop essential mental representations and the necessary technical knowledge. Educators emphasise that understanding the concepts of programming are more valuable and language-agnostic than the language itself. Learning these less-popular languages ensures that students develop the mental representations necessary for programming.

“A degree in computer science isn’t a vocational course on programming. Learning a programming language is not the object for the exercise. Understanding what programming is and how to solve the problem computationally, such that when you’re thrown into a new language and a new situation, you can get your bearings with reference to some broader concepts.” – E1

By giving students the cognitive foundations required, students are expected to reach a point where they can translate a real-world problem into computational space, allowing for it to be solved through software code. This is a critical milestone as it ensures students can successfully engage in more complex software engineering tasks.

“The purpose is to try and get them all to a level where they can think about a problem as a computer programme.” – D1

Staff need to ensure students experience the opportunity to solve problems independently and own the responsibility for developing their understanding. Part of this means staff should be careful not to obstruct

students' learning opportunities by resolving issues that are or just above the level of where students should be.

“One of the things I found really hard is to tell my teaching assistants to never solve the student’s problems, Definitely not on their keyboard, unless it’s something weird. But in general, just to say, What, the problem here? And to engage [students] in this process” – A4

A problem solving mindset can be facilitated using projects that necessitate critical approaches. Projects often require a mix of both technical and non-technical competencies, which can be used to demonstrate to students the value of soft skills. Project-based learning offers opportunities for students to apply the more abstract understandings developed during puzzles or low-level language-based tasks with more relevance to software engineering.

“[The] programming project is not an easy module, and it is made deliberately so that students learn how real world applications work in terms of testing and debugging. At an advanced level, C is not easy to work when you have to do a lot of memory allocation and deallocation and most of the students struggle with segmentation faults. They don’t know where the code is going wrong. So it’s basically approaching the problem solving skills and following good programming practice.” – D3

The progression of students' cognitive models of software engineering often begins with removing software nuances (using logic puzzles or low-level languages) before exposing students to increasingly complex situations. Once they possess the necessary mental models, they can perform tasks related to software engineering careers.

“When you ask students to find the one line that has a bug on it in a 6,000 class Java programme, they have to be analytical, and we teach them the code reading skills to help them track down the source of the bug.” – A3

The two previous comments succinctly highlight what the end goal of a student's cognitive development is driving towards – the ability to effectively problem solve using a critical mindset. Critical thinking can be defined as “being able to critically review the information and determine its importance for informing decisions and being able to use evaluative and inferential reasoning to increase the likelihood of a desired outcome” [34], and this is what educators are encouraging students to develop. Through an incremental development process, students are directed towards these essential cognitive capabilities required for technical tasks such as programming.

Social. Students are encouraged to develop their communication and group working skills during their education. Communication includes aspects of speaking, reading, and writing. Group working skills focus on how communication is used in intragroup and intergroup contexts to complete shared goals. The pedagogical aim of social development is to help students communicate in professional and articulate ways. Peer discussions are valuable for developing communication skills and boosting students' understanding. By allowing multiple perspectives to be presented, students can challenge their preconceptions by reflecting on other perspectives. Discussions can be incorporated through software engineering relevant ways using asynchronous tools. The tools mentioned include institutional education systems (e.g., Moodle and Blackboard) and industry-specific tools like GitLab. Asynchronous tools allow educators to help facilitate discussions, and they provide features that promote more efficient working (such as notifications) and practical applications to future employment.

"We adopt Gitlab, but we use the bug trackers as generalised issue trackers. We tell them to use that issue tracker as a general task planning thing. So put deadlines and assign them to people. If you've got issues, put comments in there so notifications get shot round to people. Because that's the way it's going to be in a company." – B2

Two skills mentioned together were reading and writing. They form a valuable part of a student's skillset, spanning technical and academic materials. These skills are typically not taught explicitly but are indirectly required to excel academically.

"I think reading comprehension is a soft skill. I would say we don't teach it. We assume they have it and that's an interesting thing." – C2

The ability to actively comprehend written materials is also an important aspect of student development, allowing them to explore information critically to further their technical skills. Reading skills can be facilitated by engaging students in various materials, including academic and technical information, and teaching them about relevant or trustworthy sources.

"I show them there's quite a lot of reference material and if they want to know how a function works, they look up the definition of it. If they want to use a library function in C, it's really well defined exactly what the inputs are, what the output is, exactly what it does. If you give it the wrong inputs and if you can read that and understand it, then you can use that function and be happy." – D1

Students must understand how to utilise communication in both intragroup and intergroup contexts. Successful groups consist of individuals who share common values or goals, which ties them into a group structure; where these values are not shared, it can result in group breakdown that impedes goal-directed behaviour. Students must learn about how groupwork influences different tasks and situations, and one skill that comes with groupwork is the flexibility of approach. A2 facilitates this by giving students various tasks, and by experiencing group successes and difficulties, they provide opportunities for students to develop an understanding of group functionality.

"Just because working in pairs as part of your team worked for task A. It doesn't mean it's necessarily going to be the perfect one for task B. For that first session we let them just go for it. And we'd rather that they, through trial and error, figure things out and naturally learn." – A2

Groupwork does not necessitate in-person working. Asynchronous working styles were emphasised by educators for communication, and it is also necessary for students to develop remote group working skills, as it reflects how tasks are typically handled in employment. Communication between group members (intragroup) can be facilitated on a course cohort level, creating a shared identity for all students on the academic course. Through systems like Blackboard, students can request and provide assistance, allowing them to explore how communication is used within intragroup settings to achieve shared goals.

"I think the soft skill they learn is how to collaborate with different members of the team more efficiently. We have a space on our Blackboard system where if they have any questions they can ask them. I encourage other students to chime in and try to answer them. I monitor these discussions and if I see that an answer is not correct, or it needs adjustment then I can jump in. They create a self-organised community where they're trying to help each other in answering questions" – A1

Longer-term projects offer students team working experiences to prepare them for future careers. Many educators use group-based projects as they combine multiple soft skills and give students experience in more

complex, long-term tasks. Projects typically require the distribution of specific tasks because the project is too time-intensive for one person to complete.

“The goal of it is to give students experience of doing a medium scale project from: ‘here’s an idea’, coming up with requirements, a design, implementation, testing, it being done for an actual end user as opposed to themselves. As well as doing it in the context of a group because most software development in the future will happen in groups.” – B2

Conflict Management exists as a by-product of teamwork because students will invariably experience group breakdown. A3 highlights this issue of working collaboratively via GitLab and where differences in working style can lead to frustration. Staff consider it a learning experience in understanding effective collaboration and different working styles.

“When they have a team member who just does nothing and is invisible to them. They say, we’re all pushing our commits and updating our issues. But this person, we have no idea what they’re doing. And sometimes those people will come in the day before the deadline and push a lot of stuff. So from that they start to understand the important role of visibility of working and managing a team.” – A3

Group breakdown can occur when students distance themselves from others and fail to contribute. In these situations, emotions can be high, and it can be difficult to articulate or justify complaints about peers. D2 offers students a solution through practical methods, such as taking meeting minutes. These provide a way for students to articulate issues, with the benefit of educators being able to review minutes for fair grading.

“I think at the time they find [taking minutes] very helpful because almost all of them have to deal with one group member who’s not really performing. And so, learning how to deal with them through the minutes is a useful skill. It takes the characters and the personality out a bit.” – D2

Students are exposed to intragroup skills such as communication, conflict management, and handling group breakdown, as well as intergroup skills, including communicating with individuals outside their project groups. Some courses introduced components where students were expected to communicate with various stakeholders. Intergroup communication (between groups) requires a different set of skills than intragroup communication because individuals may not share the same goals or values, requiring different means of communication. Students experienced intergroup settings where they were required to engage with “clients” (from within the university) as part of requirements gathering. This intergroup working was not a common approach to education, partly due to the complexity of arranging it.

“It’s about talking to the client and getting the requirements out of it and learning the important skill that the client isn’t technical and has no idea what they want. And how do you handle that relationship?” – E1

Educators aim to inculcate students about the relevance of intergroup communication by offering workplace context to specific skill development. This highlights that more than technical skills are needed, particularly when others possess the same technical skills; students must understand how to engage with management or stakeholders to communicate a point effectively.

“You can be very good at coding but if someone else is better than you in communicating what they have done, what are your unique skills? What is the unique aspect of your work then if the other person can do the job. So I think this kind of argument for them, this is very important, and it catches their attention.” – B1

Intergroup communication was much more limited compared to intragroup exposure. However, intergroup communication can also be framed through the sustained interaction between educator and student, which

manifests through presentation opportunities or assessments where students are required to deliver high-level information to an individual outside of their group. These development opportunities are much more implicit and speak to the nature of the education system.

Professional Skills. Professional skills, which includes ethical thinking and time management, are valuable for successful employment. In SE2014, professional skills include ethical thinking, whereas time management is not referenced, indicating the implicit expectation to manage and complete activities across the curriculum.

Ethical thinking refers to the way people consider the implications of their actions. It is commonly developed in association with communication, allowing students to engage with each other and be exposed to a greater range of perspectives, which educators can mediate.

"We asked students in groups to read 10 positive news stories and 10 negative ones, and to make some short notes about why they think this technology is positive, why they think this technology is negative. And at the end of the session we asked them to reflect on the ethical implication, social implication, economics of industry, implication of what they have read. I think this is kind of connected with the 'what does it mean to you', to reflect on the impact that your work can have in real life" – B1

Others mentioned opportunities to employ educators who specialise in topics such as ethics. In doing so, they gain the expertise, passion, and interest for incorporating ethics. One of the interviewers noted that not all modules were necessarily taught by domain experts (and whilst they were referring to technical experts, the notion stands for soft skills) who sometimes lacked the passion required. This may transfer to the development of students' valuation of these skills, as they typically converge towards staff values [38].

"Ethics spans right across the university and they set up a teaching centre where they could collect appropriate staff and those. Staff get seconded across various courses and there's also a faculty ethics team as well that looks at issues across our faculty. If there's an ethical component to what we're teaching, we would generally second someone from that unit to teach it." – D1

Time management is recognised to be a complex skill to teach, but that it is essential for students to learn. Looking at the course holistically, A1 notes that the department offers so many activities that it is unrealistic for a student to attend everything. From this, students must manage their time to attend to what they consider the most important things for their personal goals.

"If they attend every single class, lab, lecture, and all the external activities that we organise, there is really no time to go home and do exercises. So they have to make decisions of what to attend and what not to attend." – A1

Students must manage their time successfully and know when to finish an open-ended task. The diminishing returns on programming tasks can be meaningful learning opportunities. Programming assignments can be made intentionally challenging and time-consuming. However, these attributes offer pedagogical advantages, requiring students to effectively manage their time and decide when they have sufficiently completed an assignment.

"A lot of interviews have coding interviews, and they use similar style principles. So they have five days to do it. It's hard. It takes time. We had a few students, who said 'I just didn't even bother with the final task', but that's fine. That's the point in a way, as well as to know when to manage their expectations, we have some students who will stay up till who knows what hour in the morning trying to hammer this bit of coursework to gain an extra 1%." – A2

3.2.2 Perceptions of student engagement. Student engagement was primarily negatively perceived by educators. Staff felt that students misrepresent the purpose of a CS degree, limiting their ability to develop soft skills. These misconceptions lead students to attend to their technical development and reduce their attention towards soft skills, resulting in students avoiding or rejecting opportunities to develop or use soft skills, as they perceive them as irrelevant to their careers. Subsequently, students experience a temporal delay between the opportunities to develop soft skills and requiring them, highlighting the event-driven nature of soft skill development. This mismatch may contribute to the skill gap observed between education and employment, where students are not expecting to be exposed to soft skills content.

Preconceptions about what a CS course encapsulates are tied to how modern university education is perceived. As university fees increase, students demand greater value for their money, representing the “student consumer” [53]. This consumer identity may legitimise student expectations that educators must teach the technical content students want to be taught, rather than soft skills. This transactional relationship means students evaluate the cost-benefit of their education, and where they perceive no benefit, they challenge why they are being taught these skills. As students are seen to devalue soft skills, this can lead to tensions between what students want and what they would benefit from being taught in the long term.

“Getting people to sit still and listen when they think they’ve signed up for something, and that’s the issue. A lot of the current mood is this notion of we are the customers, now give us what we want.” – E1

Student preconceptions and previous experiences can influence how they internalise the soft skills material they are presented. Naïve perceptions of what is required to secure a career may lead to an overvaluation of technical skills, either because of prior experiences (such as programming hobbies) or a misunderstanding of software engineering.

“They’re coming back to Uni after maybe some gap year, and they don’t see the purpose of [soft skills]. The students who just arrived from school, they feel [soft skills] don’t have anything to contribute from the career perspective” – C1

Students may misrepresent what a software engineering career involves, expecting a highly technical endeavour, reducing their receptiveness towards soft skills. Students may avoid engaging in soft skills because of these perceptions, contributing to the soft skills gap. This finding points towards more general perceptions of students, in that underdeveloped skills may remain this way because students refuse to engage in difficult tasks.

“I have guest speakers come and talk about what it’s really like working in industry. One of the students asked, ‘what if you don’t like working with other people?’ And the industry speaker said, ‘please don’t work in the software industry.’” – D2

“Sometimes it is hard to convince our students that we are not making their life difficult.” – D3

Students may refuse soft skills because they devalue any skills associated with tasks they find challenging and avoid both the task and similar tasks that involve the same skills. Their reduced value results in students attending less focus on their development in favour of activities that they find more enjoyable or rewarding [37].

“I think there is a mismatch between what we do and what the students expect, or even what the students are good at.” – D1

Building off D1’s comment about a mismatch between the CS course material and what students are good at is reflected in the following quote by B1, who notes that students do not enjoy writing outside of

programming scenarios. Steady constant exposure can be essential to ensure students recognise where these skills hold value to counter disengagement with skill development. It is a gradual process, but it is necessary to emphasise the persistent presence of soft skills in software development.

“Students hate to write. They’re not used to it. And the first reaction [is] we don’t want to do that. But then this is a little bit of work that [I] do across the course to create a kind of consistency [to] make them understand why this is important.” – B1

Soft skills and their domain-agnostic nature can result in students disengaging as they do not see a direct link between the skills and software engineering, emphasising a student’s misperception of the CS degree being a programming course. The students’ categorisation of soft skills as being more relevant to other professions evidences their view of computer science and reflects a misunderstanding of what is required in the workplace, which may involve managing or leading a team as they progress through their careers.

“Various bits of peer marking go on which has to be moderated. There’s a tension that we often say it’s for you to try and bring on the rest of the group. [And students say], I didn’t sign up to do a management or a leadership course. I’m not a teacher. Why am I having to supervise this guy? And that’s working in a team.” – E1

A temporal delay was observed between the point where students are first introduced to soft skills and the point when they realise the importance or relevance of these skills. Staff perceive students to reject soft skills during their educational journey, and it is only when they enter the workplace (through placements or employment) that they recognise their value.

“What they don’t realise is what they’re buying is me knowing better than them about what they’re going to need in ten years’ time. And that’s where we come to those project weeks. They pathologically hate them until they’ve graduated, and they come back and say, ‘no, that’s the thing I talked about in an interview. That’s the thing that’s helped me for most of my career’.” – E1

The temporal delay reported reflects the event-driven nature of soft skill development. The following two quotes represent two possible event types that can bring on the valuation of soft skills. In A4’s example, they refer to students recognising the value of soft skills in situations where they were originally taught these skills against their desires, and it is post-graduation when they are required to use these skillsets that their impact is recognised. In contrast, D2 describes a different experience students undergo, where an absence of soft skills drives the recognition. Students must have been exposed to these soft skills at some point in their education, where they were required to be implemented to recognise this absence. The event should offer students opportunities to judge whether to use these skills. In choosing not to use these skills voluntarily, their absence may be recognised, driving the recognition of the need for the skill.

“so we have got a lot of anecdotal evidence where people who hated their second year software engineering Course unit two years after graduation, come back and said this was the most relevant thing I’ve ever learned.” – A4

“When I supervise my third years, I say, ‘Guys, I’m leaving it up to you to work out how you want to work in a group. You’re welcome to do all the minutes and stuff I taught you in the first year, but I’m not going to check’. Invariably, they don’t bother, but usually at the feedback at the end they say they wish they had done so.” – D2

Staff perceive the student experience of soft skills as largely negative, with students rejecting or avoiding opportunities to develop these skills. This rejection is based on student misconceptions about soft skills and

their value. Soft skill development is an event-driven process in which students must experience situations where soft skills are either used successfully and tied into goal attainment or where goals were not met due to an absence of a specific skill.

3.3 Discussion

In this first discussion, we cover the main findings and limitations from the interviews before a more comprehensive general discussion later. We reported on the staff perceptions regarding the embedded soft skills. Interviews were conducted across five institutions, offering an enhanced understanding of the soft skills gap between graduates and employment. Using a thematic analysis, we identified core soft skills, which can be separated into cognitive, social, and professional skills. We also identified a pattern in how staff perceive students, with a general refusal of soft skills during education, and only once students enter the industry do they understand the value of soft skills.

3.3.1 Staff Perceptions of how Soft Skills are Taught. We answer the first research question, “How do academic staff understand how soft skills are taught in their core modules?” by identifying the skills taught directly and indirectly. Direct methods relate to the intentional incorporation of skills in the course, and indirect methods relate to where skills are a by-product of other activities. Skills can also be delivered explicitly or implicitly, relating to how the skills are presented.

Critical thinking and problem solving are taught explicitly, using non-programming activities to facilitate the development of a critical mindset and implicitly through debugging activities. Non-programming activities are generally viewed positively [52], as they allow the development of critical thinking without requiring an understanding of programming first. Implicit methods, such as debugging, promote reasoning around failure, which can act as a catalyst for critical thinking [17]. Critical thinking and problem solving are valuable for programming-specific skills and are core components of learning effectively, as students need the necessary mental models. Part of the learning process was engaging with the tasks and having staff present to facilitate individual skill development.

Communication was taught directly, forming a key part of teaching materials in explicit and implicit approaches. Explicit activities focused on intragroup skills, such as peer discussions, and implicit methods included tools like GitLab that provided students with experience working with software engineering tools and using them collaboratively to maximise effectiveness. Communicating through these tools is a method received positively by students as it offers more accessible communication methods [20].

Online forums and discussion spaces are reported to increase student engagement through increased discussion of sharing solutions and common concerns [23]. We see that these ideas have been used successfully in practice to the benefit of students and educators, as at least two of the universities mentioned using GitLab. Others may also use this tool, but it was not mentioned in interviews.

Ethics was often directly implemented and was predominantly viewed positively by educators. Ethics was often taught in conjunction with communication, encouraging multiple viewpoints to be heard when considering ethical and legal approaches. The perceptions of ethics held by graduates emphasise a poor regard for its role in the workplace [34], despite employers recognising its importance in the workplace [43].

Skills taught indirectly included conflict management, reading, writing, and time management. Conflict management was indirectly taught through groupwork and manifested where tensions or breakdowns occurred.

This skill arose organically as a facet of group dynamics and was not manufactured by staff. Staff perceived conflict as an unavoidable consequence of collaboration but supported students navigating these difficult situations. Reading, writing, and time management are all components of general learning and academic success.

From the direct-indirect interpretation of skills, we are afforded an idea of how skills are taught during the CS undergraduate education. Many skills are taught alongside others, and technical skills or tools were leveraged to highlight the relevance of certain soft skills towards software engineering. Understanding that skills can be taught in these different styles may mean students do not necessarily recognise all instances when they are taught soft skills and may reject activities explicitly about soft skill development. The core skills of critical thinking, problem solving, communication, and teamwork support previous studies reporting similar findings from graduate populations [11, 56], curricula [25], and employment [2, 42].

Teaching implies a student audience, and so it is helpful to recognise how student engagement is perceived by staff, as this can influence how content is delivered. Students were perceived to view their education as a “vocational programming course” (E1), which has been reported previously [29]. This preconception echoes previous findings that students think non-technical skills are unnecessary for technical careers [32]. While educators and employers agree with the importance of soft skills, our findings suggest staff do not perceive students to recognise the value of soft skills, aligning with previous findings where students undervalue soft skills [10] in favour of the skills they enjoy [37].

Students were also perceived to experience a temporal delay between being taught soft skills and recognising their value. It is often only once out of education, and being exposed to situations where either a soft skill is deployed and recognised to be successful, or situations occurs where an absence of a specific soft skill is noticeable that students realise the importance and value of possessing these skills. As many of the events that drive this recognition are industry-based, students are potentially unaware of the skills they should demonstrate in interviews [5].

3.3.2 What skills are important. The second research question asked, “What importance do staff ascribe to certain soft skills, in comparison to other skills?”. From the direct-indirect interpretation of how skills are taught, certain skills are recognised as more valuable than others. For skills to be taught directly, educators make conscious decisions to incorporate activities that promote skill development, indicating their perceived value. Indirect skills are typically a by-product of the general learning cycle. The skills referenced as being directly taught are valued enough by educators to ensure they have sufficient space within the curriculum.

3.3.3 Limitations. We recognise the likelihood of a self-selection bias for this study, such that participants report favourable attitudes towards soft skills. Indeed, the four educators who declined the interview declared an absence of soft skills in their modules. Consequently, the sample is not fully representative of a teaching department. Despite missing information on what skills are potentially undervalued across curricula, the interviewees offered insights into how soft skills are perceived by educators.

The semi-structured interview approach incorporated, at times, a more conversational interaction, which may have influenced interview topics, resulting in specific skills being mentioned that would not have otherwise. The benefits outweigh potential biases, as it encouraged greater interaction and further information from the interviewee. For transparency, the anonymised transcripts are made available in the online repository.

4 STUDY 2

The first study offered important insight into the educator perspective on their modules. Through these interviews, the manner of how soft skills are taught – directly or indirectly, and implicitly or explicitly – surfaces key information about how technical skills are often structured around the non-technical skills and are used as the method to convey the soft skill content. What remains missing, is how this information relates to how soft skills are embedded with a curriculum. In this second study, we report on a natural language processing approach to identify the embedded soft skills in CS curricula, which provides a more descriptive approach to understanding how soft skills present within CS education, and in combination with the results from the first study, it answers the third research question, “How do these skills align with the information available in online courses and module information?”.

Previous studies have used manual detection methods [18, 25], but natural language processing models can expedite the identification process and offer a more uniform detection of keywords across documents. Fareri et al. [19] developed a model that detected the presence of soft skills and applied this to job profile data, finding general skills of leadership, autonomy, conflict management, and communication as highly valued across job profiles. To the authors’ knowledge, similar methods have yet to be directly applied to curriculum information.

4.1 Methodology

Data collection targeted websites of the N8 universities, which were scraped for information related to undergraduate CS courses, including course descriptions, core module information, and any publicly available course-relevant data. The academic year 22/23 was targeted because this was the same academic year the interviews were conducted, which was achieved using the Internet Archive. Scripts for scraping these datasets can be found in the OSF repository along with the scraped data: https://osf.io/25ecz/?view_only=917e6de233484d59abaac243647c89a2. The Faculty Ethics committee approved this secondary data collection and analysis.

4.1.1 Analysis. SkillNER, a skill-based Named Entity Recognition (NER) model [3], was used to extract soft skills. An NER is an algorithm specialising in identifying keywords based on probabilities and co-occurrences. SkillNER was developed from using 31,278 individual terms (336 soft skills, 29,091 hard skills, and 1,851 certifications) taken from job adverts and other employment-related materials. The analysis pipeline is available on OSF. SkillNER was implemented using Python 3.11 and R 4.2.2 for data analysis.

The dataset was passed through SkillNER. Important soft skills that were not identified by the process were modified in the text to increase extraction rates (e.g., “team” was not recognised and was changed to “teamwork”), which was carried out by data exploration. Following data annotation, the soft skills were grouped according to the skills identified in REDACTED FOR REVIEW, as many identified terms were similar (such as “collaborate” and “collaboration”, or “solving problems” and “problem solving”). Some soft skills by REDACTED FOR REVIEW were identified as hard skills by SkillNER (such as “time management”); these were redefined as soft skills. Identified terms related to the teaching and educational aspect of soft skills (such as “teaching” or “module”) were removed. Terms that had a skill probability over .60 were retained.

4.2 Results

References to soft skills are not homogeneous across institutions, and soft skills make up between 12.90% and 27.20% of all referenced skills, with a mean average of 17.45% ($SD = 4.89$). The most frequently mentioned soft skills across the combined curricula were *teamwork*, *communication*, *professional skills*, *ethical thinking*, *critical thinking*, and *problem solving* as shown in Figure 1. *Professional skills* were the only soft skill absent in the list developed in Ivory et al. [34]. Visualising the soft skills between institutions is made more explicit in Figure 2, that some universities have a more even spread across mentions of soft skills, but still, the pattern is inconsistent across institutions, and Figure 3 on an institutional level, offering complementary perspectives.

The institutions are disparate in the levels of soft skills reported. The figures indicate little agreement between institutions regarding how frequently soft skills are mentioned. These figures present the frequencies as proportions because different word counts were extracted from each institution. Teamwork has a presence across all institutions. Professional skills are dominated by three institutions (Lancaster, Leeds, and Newcastle), and most institutions refer to each of the key six skills, except an absence of ethical thinking for Sheffield and problem solving for York.

Skill co-occurrences – how often two skills appear in proximity – can be a useful way to understand skill relationships. In Figure 4, the technical and soft skills are plotted together in a network-based upon within-sentence co-occurrence. The more frequently co-occurring skills are denoted by the strength of the connecting edge, and node size is relative to skill frequency.

We observed that soft skills of *teamwork*, *communication*, *professional skills*, *ethical thinking*, *interpersonal skills*, and *critical thinking* are strongly linked. Technical skills of *software engineering* and *software development* are intrinsically linked with the cluster of soft skills, emphasising that these are taught as team-based activities. *Problem solving* was mentioned more frequently alongside *programming* skills and *autonomy*, indicating that programming emphasises problem solving skills and independent working.

Figure 5 displays relationships between the soft skills and modules across all institutions to assess whether soft skills are taught together or separately. The nodes' size indicates the number of associative links, and the most dispersed skills were *communication*, *critical thinking*, *ethical thinking*, *professional skills*, *problem solving*, and *teamwork*. It can also be seen that 38 modules are only associated with a single soft skill.

Figure 6 offers a different perspective by displaying the proportion of modules that include one of the key six soft skills. It is seen that despite teamwork being the most frequently identified skill (by term matches), it is typically present in around 20% of modules. Communication is most broadly embedded, appearing in over a third of modules in five institutions. Professional skills are widely embedded within Newcastle's curricula but less so for other institutions. The data in these two figures shows that the distribution of soft skills throughout curricula is highly dependent on institutions.

Exploring the soft skills taught across the course structure separated by year, in Figure 7, we see little pattern across the different institutions. The heterogeneity is apparent where some years contain no soft skills, suggesting an inconsistent approach to introducing and teaching soft skills. Ethical thinking is rarely included beyond the second year for any institution, despite dissertation projects requiring ethical approval for human-based research.

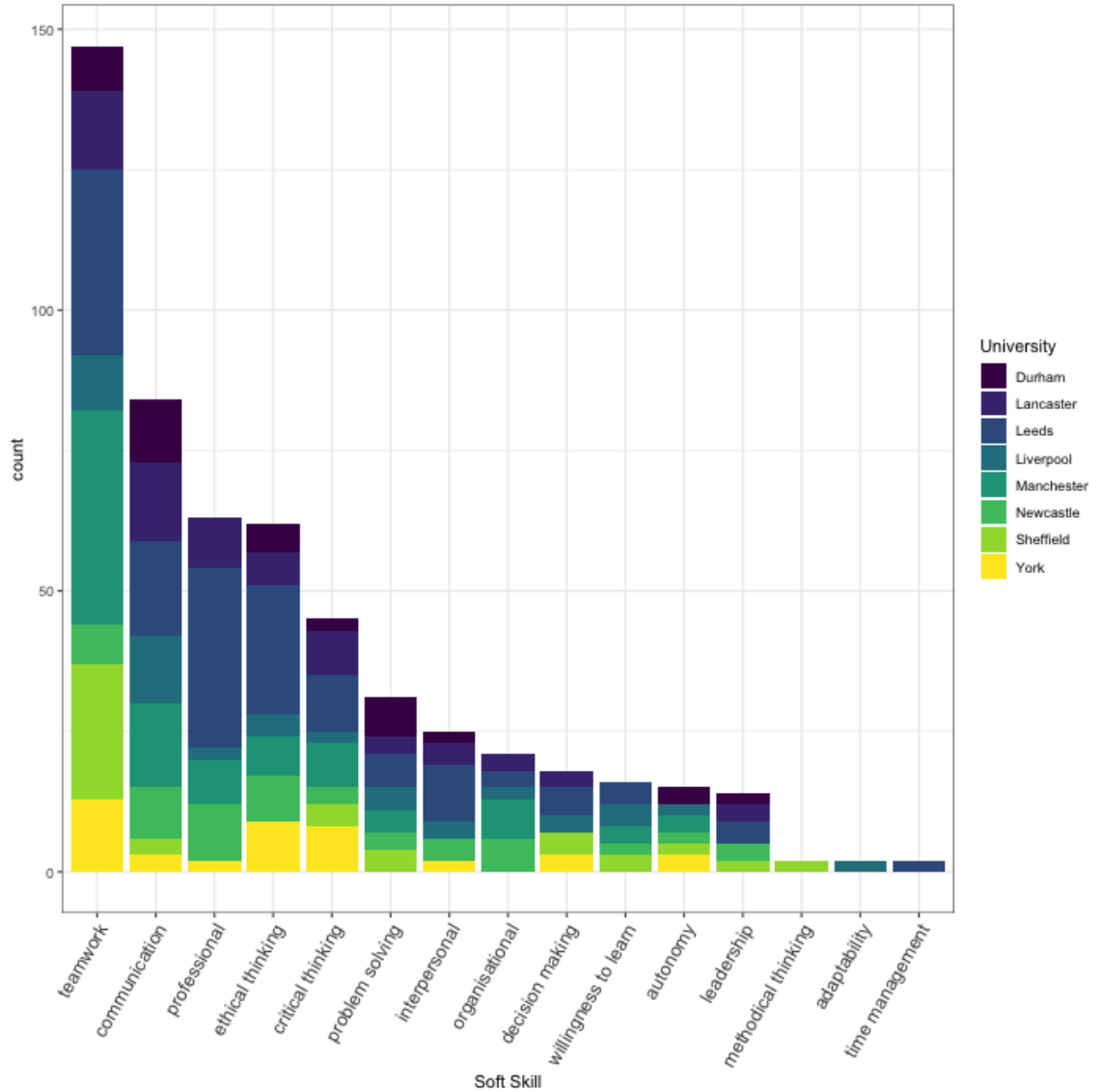


Fig. 1. A stacked column plot indicating the most mentioned soft skills across all curricula. Skills mentioned only once in an institution's curriculum were removed.

4.3 Discussion

In this curricula analysis, we extracted soft skills using an NER model. In scraping publicly available information on CS undergraduate courses, we investigated skill frequencies and co-occurrences. The main findings from this exploration are that *communication*, *critical thinking*, *ethical thinking*, *professional skills*, *problem solving*, and *teamwork* were the most frequently mentioned. No clear patterns were identified between or within institutions for how soft skills are mentioned. On an individual institutional level, institutions are

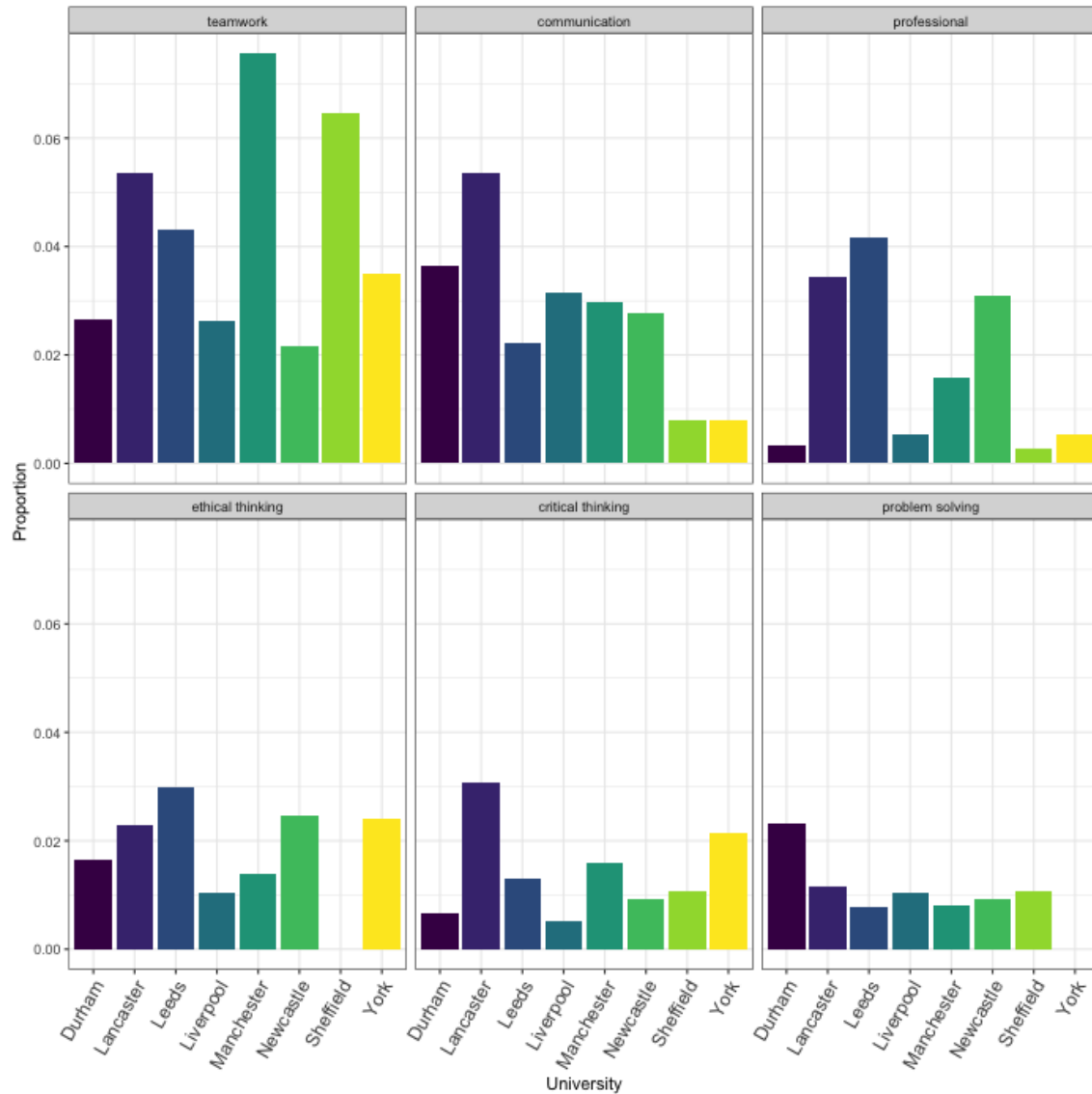


Fig. 2. Separating the frequencies by soft skill shows the patterns (or absence of) across institutions. Only the top six skills are shown.

heterogeneous in referencing soft skills, and this may not necessarily reflect reality. Some institutions may be more descriptive in their module information or reserve information for internal use, making comparisons between institutions difficult. The analysis highlights that soft skills are commonly mentioned in proximity, indicating they are taught or assessed together. The exceptions to this were *problem solving* and *autonomy*, which were more closely associated with programming. It was also seen that software engineering was closely linked to skills of teamwork and communication.

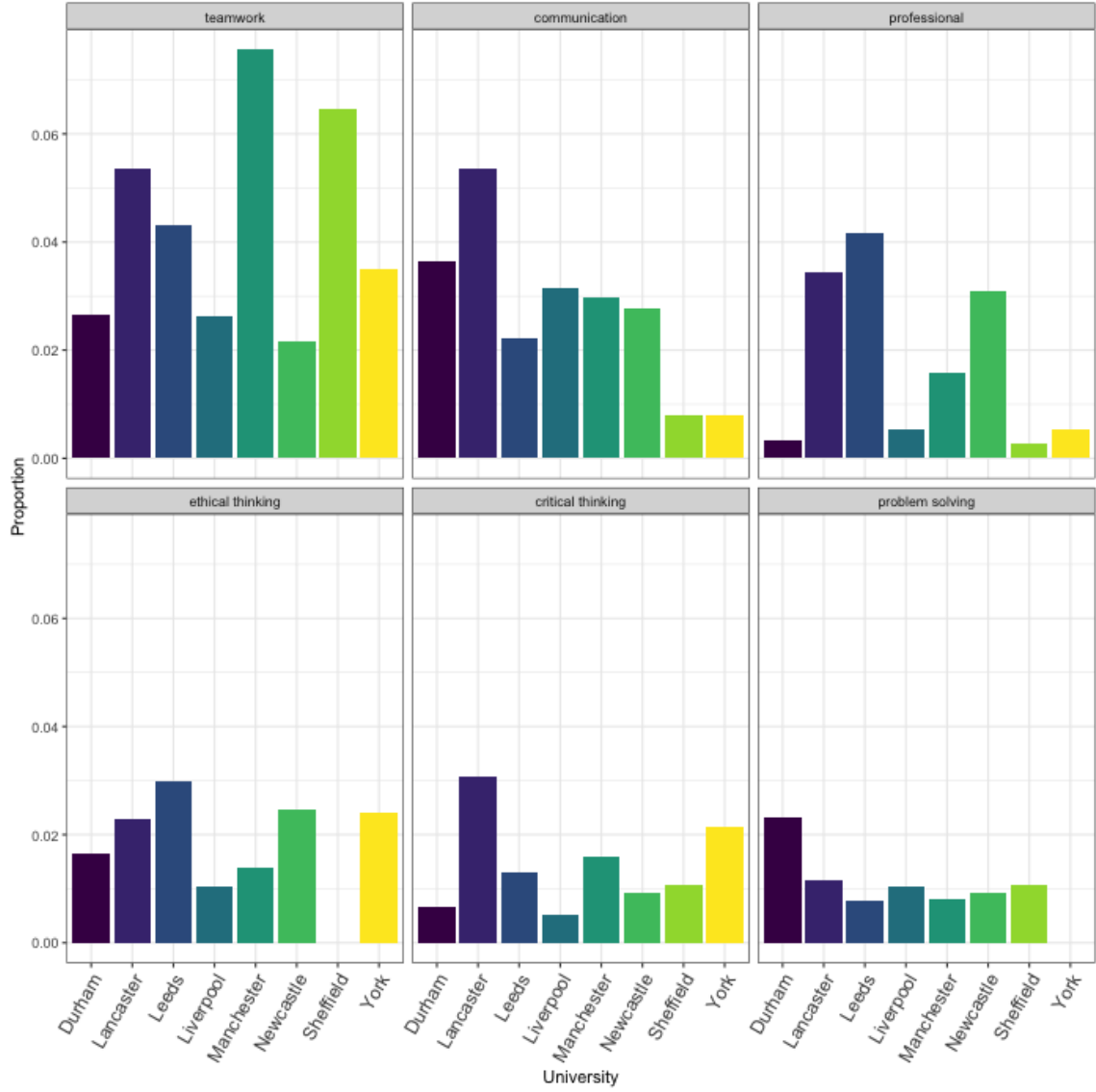


Fig. 3. Presenting soft skills split by their institution offers a different perspective, showing how institutions present different skills within their curricula.

The most frequently mentioned soft skills of *teamwork* and *communication* are those highly valued by software engineering employers [2, 35, 45, 50], as are the skills of *critical thinking* and *problem solving* [18]. This suggests higher education and industry are aligned in the skills valued, supporting the notion that CS undergraduate education is positioned to inculcate students with the appropriate soft skills for software engineering careers. This finding is further supported by previous curriculum analyses identifying *communication*, *teamwork*, and *ethical thinking* [18, 25].

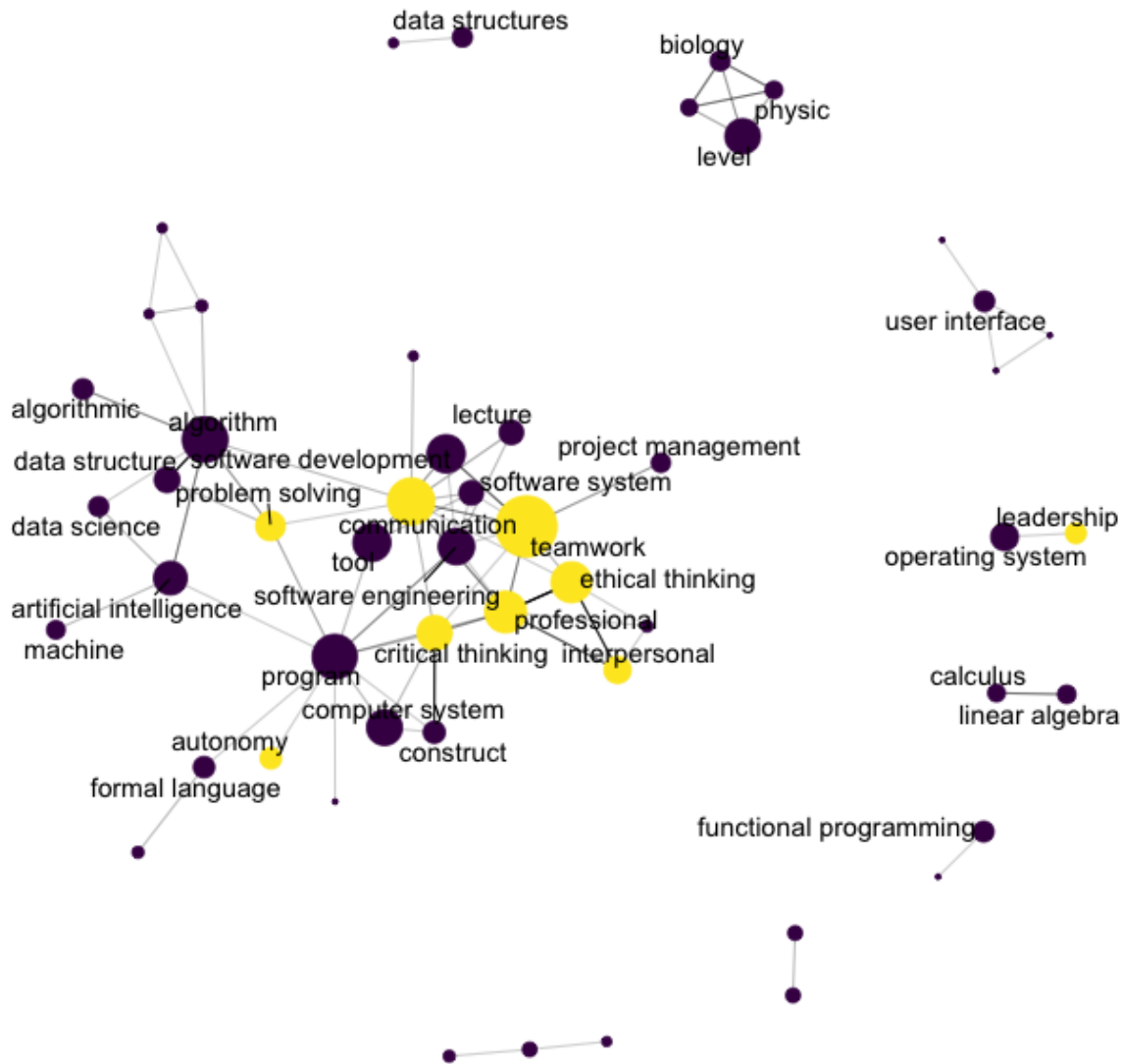


Fig. 4. Skills that occur more than three times are included, and only skills that occur more than five times are labelled. This offers a representation of how soft skills are referenced within curricula information. Terms with no contextual value, such as 'computer science' and 'module' are removed as they overwhelm the plot due to their high frequency.

Ethical thinking was the fourth most frequently mentioned skill across the curricula and saw reasonable exposure across most institutions. Despite this emphasis, graduates rank ethics very low for employment [34], and this, along with the reported soft skills gap [4], is indicative that student perceptions significantly impact how soft skills develop.



Fig. 5. Network plot of the soft skills and co-occurrences through modules. The connections indicate skills that are taught in the same modules. Yellow nodes are soft skills, and dark purple nodes are individual modules.

Skills of software engineering and software development were strongly associated with *teamwork* and *communication*, suggesting these are presented to students as social activities. This association reflects employer expectations that software engineering is carried out collaboratively [1].

4.3.1 Limitations. The NER model could not identify all references to soft skills without prior processing steps, which was addressed in the analysis pipeline, where the original text was modified to ensure that

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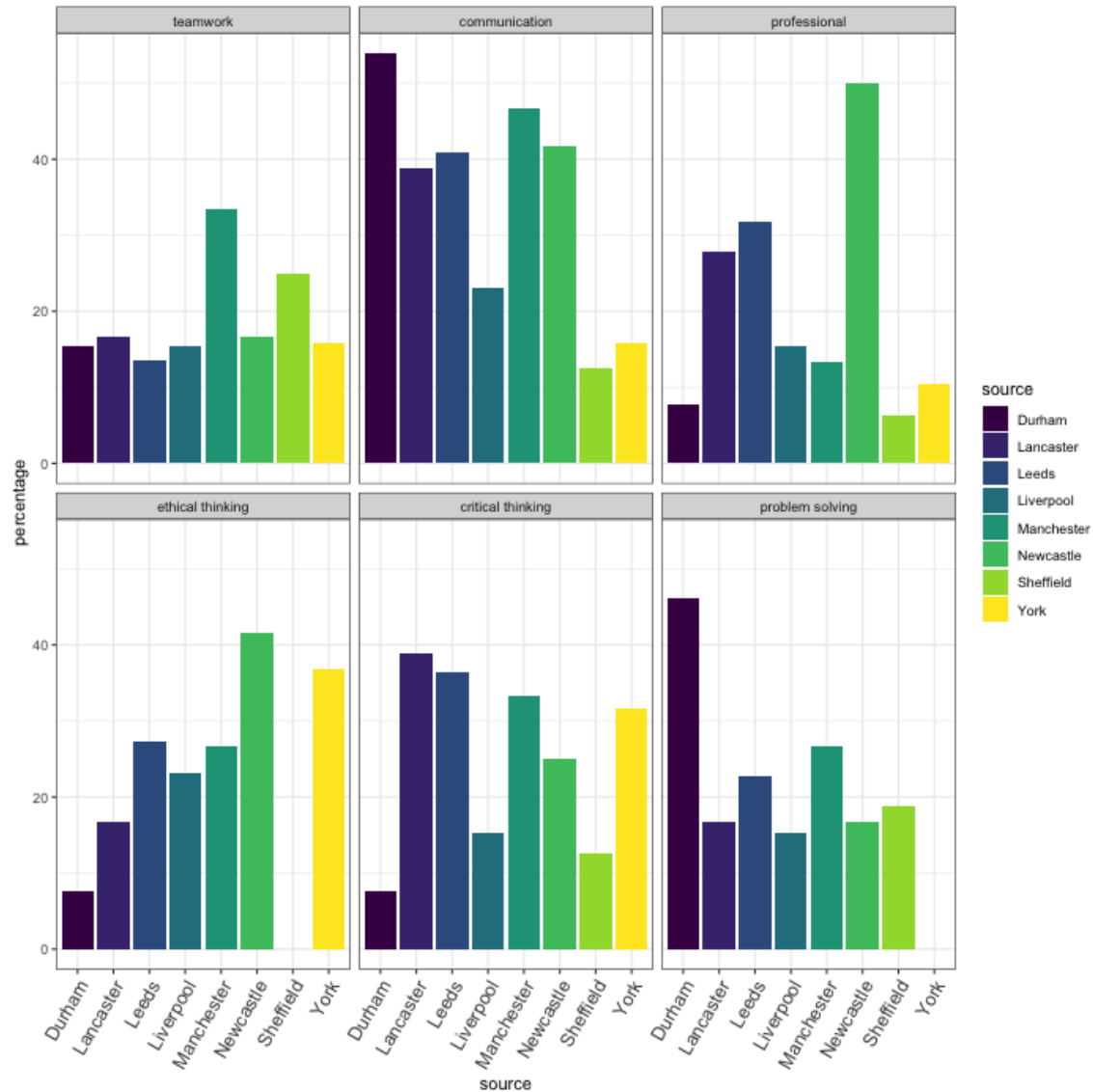


Fig. 6. Proportion of modules within an institution that includes the top six reported soft skills.

SkillNER identified references to teamwork, groupwork, ethical and moral thinking. It remains possible that other terms were not detected. Some skills, such as groupwork or time management, were classed as technical skills by the NER model, but to maintain consistency with previous research, we classified these as soft skills according to the definitions used in REDACTED FOR REVIEW . Both issues indicate a broader issue with soft skill research: they are ephemeral concepts with loose definitions. This is not just a limitation of the present study but also speaks to general soft skills research. The findings indicated reasonable alignment with current understandings of valued soft skills despite model limitations.

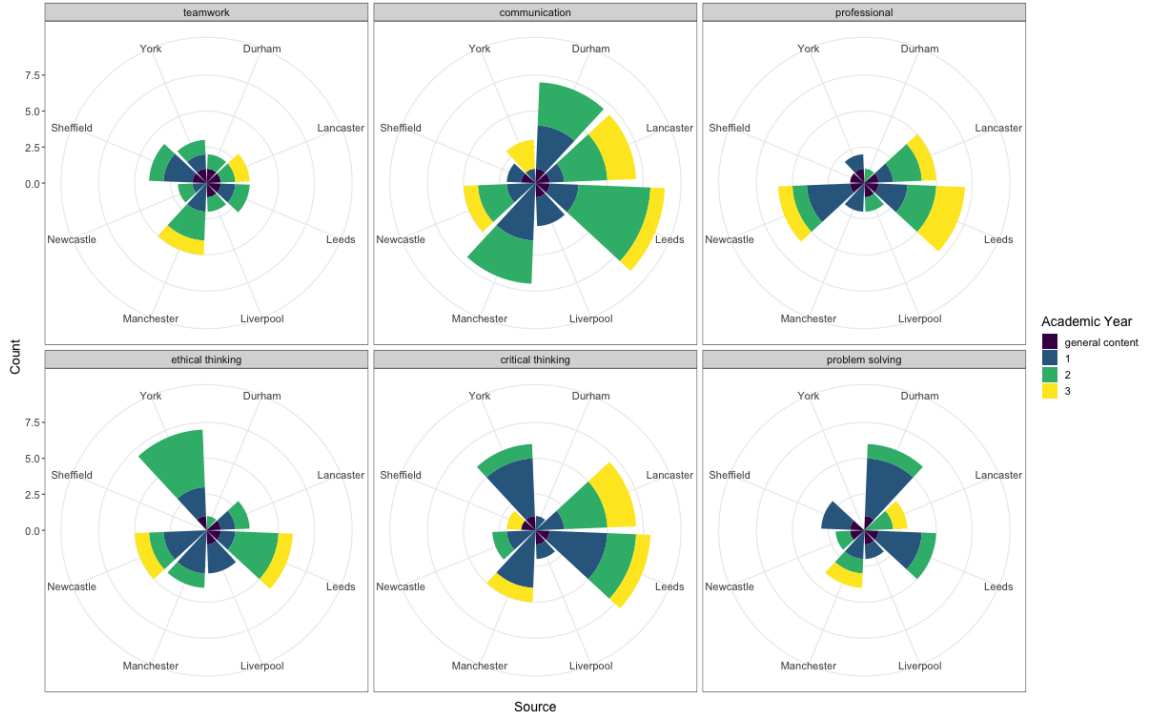


Fig. 7. Inspection of soft skills by course structure (module years), no distinct pattern is observed across skills.

5 GENERAL DISCUSSION

In this two-study paper, we first interviewed staff to identify the perceptions of embedded soft skills across core modules, followed by the extraction of soft skills from curricula to converge on valued soft skills. It was seen that staff valued skills including *communication*, *groupwork*, and *critical thinking*, which were mentioned across modules but were also extensively taught directly, highlighting perceived importance. The second study applied an NER model to core module content, highlighting heterogeneous frequencies across different institutions and a strong link between software development and groupwork. In this general discussion, we draw the findings from both studies together, and describe future research.

The first study answered research questions 1 and 2 by interpreting soft skills taught directly or indirectly. For question 1, “How do academic staff understand how soft skills are taught in their core modules?” the interviews highlighted that soft skills are taught in various ways, but combining soft skills with relevant software engineering tools or practices provides ways for students to recognise the value of soft skills. The second research question asked, “What importance do staff ascribe to certain soft skills, in comparison to other skills?”, and the skills intentionally incorporated into the teaching materials can be seen as more valuable. Staff are constrained by competing modules and time, so they must make conscious decisions about how, and which, soft skills are given space within curricula.

To answer the third research question of “How do important soft skills align with the information available in online course and module information?”, the information from study 1 and study 2 are assessed together. General agreement between the most valued skills is seen as educators spoke about *communication*, *groupwork*, and *critical thinking* as being intentionally included within their materials, and these are also the skills identified most frequently. The course information and curricula are assumed to reflect the skills embedded within the course. Extracting soft skills presents a course-wide representation of the skills that educators value. This alignment indicates that the curriculum information is accurate, and educators are teaching the soft skills they advertise.

No discernible patterns were seen between or within institutions, suggesting that despite frameworks like SE2104, there is little conformity between institutional curriculum design. This is positive in that it highlights unique institutional offerings that can be tailored to their branding, but this absence of uniformity suggests the frameworks are interpreted very differently, to the extent that certain skills are not focused on in certain institutions. This can lead to potential choices that students are required to make in their selection of education at a point in time where their misconceptions may draw them to programmes that reference lower levels of soft skills. It should be encouraged that curricula reflect the true contents of their course and ensure that the relevant skills are targeted and mentioned.

The skills identified are some of the most desired by employers, who desire employees who can communicate and collaborate effectively [2, 41, 45]. Our findings suggest educators are aligned with employer expectations, highlighting that the educators interviewed are aware of which skills are needed to secure employment. This alignment indicates that the soft skills gap may result from student perceptions and how they develop these skills. While staff perceptions influence how students perceive soft skills [10], it is reliant on students to fully realise their value [24].

Many students start their undergraduate education believing a CS degree is primarily a vocational programming course [29], accompanied by the misconception that programming is all that is required for a successful career. Despite these preconceptions, students graduate from their courses with a stronger sense of the breadth of skills needed [38]. Our findings suggest the CS undergraduate course is set up to ensure graduates possess the necessary skills for software engineering, but the difficulty is ensuring students recognise this.

The research implications suggest educators should bring forward the events that motivate students' recognition of soft skills and reduce the temporal gap between being taught and appreciating their relevance. To close the soft skills gap, students must reflect on their own goals and recognise the role of soft skills [13]. Students' preconceptions about the importance of soft skills manifest in technical skill prioritisation over soft skills [33]. While preconceptions can be altered through exposure to soft skills, bringing perceived value up for specific skills [10], this is still limited by internal understanding of their value.

Staff recognise that students fail to value soft skills unless prompted by some event involving soft skill use, so incorporating more industry-specific events into the curriculum can increase students' value of soft skills. There are multiple ways to achieve this; one method would involve industry partners offering industry experiences to contextualise soft skills. Another opportunity could be to integrate short-term industry placements into the curriculum, allowing students to experience real-world situations that offer events to reduce the perceived temporal distance. The event-driven nature of soft skill development should

be integrated into modules through project work, along with ensuring students are aware of the relevance of soft skills [36].

Further implications result from the application of the NER model. Similar approaches can be applied during curriculum designs to ensure that soft skills are embedded throughout the content. Embedding soft skills across core modules would offer repeated exposure to these skills and emphasise their alignment with technical skills. By deploying NER models, staff can ensure that soft skills are suitably distributed across modules and are not localised to specific modules.

Our research offers a unique contribution to the understanding of CS higher education and its role in teaching soft skills. Previous work has explored educator perceptions and what they teach, highlighting that CS courses should prepare students for employment [55] by giving them experience with relevant tools and methods widely used in software engineering [14]. We contribute by exploring educator perceptions about the embedded soft skills and their presence in curricula. We find that skills align with employer expectations, suggesting that students may be the weak link in the soft skill development process.

5.0.1 Future Work. Staff reportedly say students reject soft skills until they experience an event that emphasises soft skill relevance. We suggest the soft skills gap observed is formed through students’ misunderstanding of the skills’ value for employment. By conducting longitudinal studies following students through their academic careers, it may be possible to map soft skill development to their education. Using survey studies following whole cohorts and interviews to pinpoint specific moments or modules that altered a perception, it may be possible to contribute a greater understanding of how soft skills develop and potential reasons for refusal, misunderstanding, or acceptance.

One study limitation noted was the self-selecting sample. Capturing information from across all educators would offer meaningful insight into the distribution of educators’ perceptions of soft skills and where these skills exist within a curriculum. It was recognised that not all educators see soft skills as having space within their modules, but gathering representative views is necessary to understand how students are exposed to soft skills.

The NER algorithm was applied to a geographically restricted population. Exploring a larger sample and systematically mapping modules between universities can be used to investigate whether soft skills are taught in similar contexts. This development would offer important information about where students can develop soft skills, complementing existing research. All the future studies detailed here offer significant implications for CS curricula design, ensuring soft skills are effectively embedded and calibrated to allow students to recognise the value of soft skills.

6 CONCLUSION

In this two-part preregistered study, we explored the staff perceptions and curriculum presentations of embedded soft skills within the CS undergraduate course. Staff perceived soft skills of *communication*, *critical thinking*, *ethical thinking*, *problem solving*, and *teamwork* as valuable soft skills that warrant including and teaching directly to students. Staff talk about the intentional space made in their modules to teach and develop these skills whilst supporting students. Staff views on student engagement were largely negative, with students rejecting the value of soft skills during education but recognising it once in industry, highlighting the event-driven nature of soft skill development. The NER analysis of the curricula highlighted important soft

skills of *communication*, *critical thinking*, *ethical thinking*, *professional skills*, *problem solving*, and *teamwork* showing high association between staff perceptions and the descriptions of the courses. The implications of these findings highlight congruency between employer expectations and the educational course in prioritising specific soft skills. The findings imply that student perceptions likely have a strong influence over their internalisation of soft skills, which results in the reported skill gap. Our findings indicate that the critical factor in causing the soft skills gap may be student perceptions, as we observe a positive relationship between the soft skills described in curricula, the soft skills prioritised by educators, and the skills considered essential in employment.

APPENDIX

Appendix A

Module Leader and Author Interview Guide

Questions (main questions in bold, follow up questions in italics)

- Tell me about the undergraduate module that you teach. What do you think works well with your course? *What key concepts are taught to students? How is the course delivered?*
- What do you understand the term “soft skills” to mean? *< I can then bring up the definitions of soft skills that I refer to; things such as teamwork or responsibility, problem solving Some of these which are more present than we may think originally >*
- How important do you think these skills to be in Computer Science or Software Development? *Who is responsible for ensuring software developers possess the non-technical skills necessary for industry?*
- **Do you teach or emphasise any specific soft skills in the module, either intentionally or as a result of other aims/goals/external requirements of the module?** *If YES: why are these included in the module? If NOT INTENTIONAL: is there any reason why they are not purposely included? Are these the responsibility of someone else or do the opportunities for these soft skills develop naturally?*
- **What other aspects of the undergraduate programme are you aware of that focus on soft skills and how do you think these contribute to students’ development?**
- **Has the module, under your supervision, been altered or adapted to emphasise or reduce focus on particular soft skills?** *If YES: why has this been altered? If NO: why not? Do you think the balance of teaching requires no further development or inspection?*
- **What soft skills do you prioritise in teaching in the undergraduate programme?** *Do you think this is different to what students prioritise?*
- *Why are they prioritising these skills? How do these skills integrate in the course? Are they introduced and students left to their own development, or is it more intentional and they’re pushed to improve? Contentious perhaps, but do these skills help improve students’ career prospects or are they more ideological?*
- **Do you have any further thoughts or ideas relating to soft skills in software development and computer science?** *Anything that has not been covered by previous questions? Any last thoughts on the balance of technical and non-technical skills.*

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